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The challenges of planning language objectives in content-based ESL instruction

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Abstract

The purpose of this research was to investigate the major patterns in content-based instruction (CBI) lesson plans among practicum teachers at the final stage of an MA TESOL program. One hundred and seven lesson plans were coded according to a typology developed to evaluate clarity and identify areas of potential difficulty in the design of content and language objectives by TESOL teacher candidates for use in PreK–12 ESL classrooms. Participants in our study tended to have more difficulty in designing language objectives than content objectives. There was also a tendency to write language objectives that focused heavily on the four language skills and on vocabulary, and considerably less often on grammatical structures, functions, or language learning strategies.

Keywords

Content-based instruction, language objectives, lesson planning, teacher language awareness

I Introduction

While integrated content and language instruction has been shown to be a highly effective means of developing second language proficiency, it is also clear that this integration is a complex pedagogical challenge, requiring robust attention to language as well as discipline-specific learning (Dalton-Puffer, 2007; Lyster, 2007; Pica, 2008). 'Content-based instruction' (CBI), as it is referred to in the USA, incorporates a wide spectrum of

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'instructional approaches that make a dual, though not necessarily equal, commitment to language and content-learning objectives' (Stoller, 2008, p. 59). On one end of the content/language-driven curriculum (Met, 1999), 'incidental', rather than intentional language teaching, results in teachers inadvertently prioritizing content over language learning (Lyster, 2007) and, on the other end, foregrounding language goals at the expense of meaningful content learning (Cammarata, 2009).

In the US PreK–12 English as a second language (ESL) context, classroom teachers provide the bulk of instruction for English language learners (ELLs) with little to no preparation in second language (L2) teaching (Gándara, Maxwell-Jolly, & Driscoll, 2005; Menken & Antunez, 2001). In these settings, ESL teachers may be called upon to work collaboratively with their content and classroom teacher counterparts in order to co-plan integrated content and language instruction. However, we know that collaboration between mainstream and ESL specialists, which could potentially provide a framework for planning language and content goals, faces tremendous barriers, with the scales tipping consistently towards content-area rather than language learning goals (Arkoudis, 2006; Creese, 2006). How well prepared is the ESL teacher for planning language instruction when this time comes? As teacher educators, we have been troubled by the question of whether, indeed, our teacher candidates were ready to meet the challenge of planning for language instruction in CBI. Our experience has shown us that ESL teachers often still struggle with addressing their students' language learning needs in content-based environments, as evidenced by their lesson plans and observed instruction. The development of learning objectives as part of lesson planning is one way of identifying what the focus of a lesson should be; by analysing teacher candidates' stated language learning objectives, we should be able to identify at least part of what they intend their lessons to achieve. While the ability to articulate clear objectives may be a separate skill from actually delivering the instruction and assessment needed to accompany these objectives, still, we consider the organization and articulation of language-focused instruction to be a critical developmental skill for educators of ELLs. If teachers are not intentionally organizing instruction focused on language learning, how, then, are ELLs provided with instruction that actually targets their language development needs?

While there has been research on both general and ESL teachers' lesson planning processes, there are only a few data-based studies of the lesson plans themselves that ask questions about what the tendencies of ESL teachers actually are when designing language learning objectives. The purpose of our research was to investigate the major patterns in CBI lesson plans among practicum teachers at the final stage of an MA TESOL program. We hope, from this research, to contribute to TESOL teacher preparation and the PreK–12 learning of ELLs by using the results of this analysis to identify and present the specific areas in need of greater attention when preparing novice teachers to develop language objectives in content-based instruction. Our research questions were:

1. What are major characteristics of language objectives embedded in content area lesson plans, as written by TESOL teacher candidates?
2. What are the challenges for TESOL teacher candidates in writing content and language objectives?

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II Review of the literature

1 Lesson planning as pedagogical reasoning

Lesson planning reflects teachers' prior experiences as learners and teachers, knowledge of their class of pupils, pedagogical content knowledge, and theories about learning, in a visualization act of an imagined future lesson (Ho, 1995; John, 1991, 1994; Warren, 2000). This entire process may result, of course, in a lesson that differs from the plan, yet it is the planning, and the plan itself, that exposes teachers' beliefs, understandings, and orientations.

Internal factors such as teachers' beliefs about students' abilities, interests, and participation, how the subject ought to be taught, and the role they wish to play in the classroom all influence the choices made in lesson planning (Bullough, 1992; Yildirim, 2003). Approaches to the act of planning have been shown to range from a linear, Tylerian style to more iterative processes. Differences have been noted in the manner in which teachers from different content areas, experience levels, or grades conduct their planning, in terms both of process and emphasis (Sardo-Brown, 1996). For instance, Kagan and Tippins (1992) found elementary teachers to focus more on the learning activity than the content goal. Additionally, they noted that elementary teachers wrote progressively briefer and less detailed lesson plans over the course of a semester than secondary teachers, whose plans became more and more detailed. In contrast to experienced teachers, novices often derive their plans from scripted guides, textbooks, or pre-packaged curricula (Westerman, 1991) with a dominant concern for classroom management, and address planning on the daily or 'survival' level (Dunn & Shriner, 1999).

A particular challenge for those involved in pre-service teacher lesson planning is the difficulty of synthesizing these internal and external factors and making choices about how to organize the lesson plan, with little classroom teaching experience to call upon (John, 2006; Mutton, Hagger, & Burn, 2011). Additionally, as teachers are much more likely to consider the appeal of their activities rather than base their planning around learning objectives (McCutcheon & Milner, 2002; Wiggins & McTighe, 2005), learning how to craft objectives is particularly challenging.

2 Teacher language awareness in instructional planning

Lesson planning in the context of PreK–12 TESOL teacher preparation is unique in that candidates are asked to design instruction targeted to develop content-area understandings (mathematics, science, social studies, literature/English language arts), while simultaneously developing English language skills for the non-native speaker.

> One of the major challenges for English language teachers is that the medium of instruction is also the object of instruction. Therefore, the lesson itself comprises the lesson activities, instructions for carrying out those activities, and the language required to engage in those activities. (Liyanage & Bartlett, 2010, p. 1363)

Attention to language instruction, especially academic language, within the content areas is understood to be the desired means to support ELLs, yet is also one of its most sophisticated pedagogical challenges (Met, 1991; Stoller, 2004).

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In CBI, the ESL teacher is expected to be able not only to shelter content, through assessing the linguistic demands of content-area materials and providing necessary textual, structural, and discourse adaptations to make that material accessible to ELLs, but also to set language learning objectives. Many times, content and language goals, however, may seem competitive, rather than complementary, with 'teachers coming to the conclusion that there is no time to teach language and content because each follows a parallel track' (Bigelow, 2010, p. 37). Without careful pre-active planning for language instruction, ESL teachers tend to grant preference to content learning goals over language teaching (Mackey, Polio & McDonough, 2004; Short, 2002; Stoller & Grabe, 1997).

In order to truly intertwine abilities – to shelter content and to teach language – teachers need a high degree of language awareness, known in the L2 cognition research literature as the construct of teacher language awareness (TLA) (Andrews, 2007). First, TLA encompasses knowledge about language in the form of a teacher's grasp of language systems and structures (metalinguistic awareness), or their knowledge about language (KAL) (Williamson & Hardman, 1995), referred to by Shulman (1986) as content knowledge. 'TLA incorporates a procedural as well as a declarative dimension, with knowledge of subject matter (i.e. the language systems knowledge base) at the core of the declarative dimension' (Andrews, 2007, p. 31). In addition, TLA includes a level of empathy for and understanding of ELLs in order to select appropriate methods and strategies for teaching language (Andrews, 2003). This is akin to Shulman's concept of pedagogical content knowledge (PCK).

Preservice teachers develop these domains of TLA by first developing their conceptual knowledge through their coursework, and then learning to apply this knowledge to their teaching through the practice teaching/practicum experience provided in the TESOL program of study, and are often assessed through lesson plans as well as classroom observations. Research on L2 teacher cognition has addressed the connection between teachers' beliefs about language learning and their actual teaching. However, there is less research on the ways TLA begins to manifest itself in the lesson planning stage of ESL teachers. The research that has been conducted shows some indication that even without the extra work of attending to content learning in a language classroom, novice ESL teachers focus more on classroom management issues than language teaching objectives in their planning (Bailey, 1996; Nunan, 1992; Richards & Pennington, 1998; Woods, 1996). Adding to the cognitive challenge of calling upon TLA in the planning process, CBI teachers must also address subject matter content and PCK for teaching this content. Therefore, the CBI teacher must learn to utilize knowledge of language and knowledge of subject matter, alongside PCK for language and subject matter teaching. In TESOL preparation programs, these multiple competencies are developed throughout teacher candidates' assignments and practicum teaching experiences.

TESOL teacher educators' research into the practices and beliefs of their candidates has noted teachers' difficulty in making 'overt connections between declarative and procedural knowledge, between KAL and the classroom' (Popko, 2005, p. 402). In Popko's (2005) case studies of TESOL candidates' planning and teaching, teachers were unable to relate their instructional decisions to their language knowledge, drawing on prior personal experiences as students or as teachers rather than on their TESOL coursework. His recommendations included making more explicit connections to TLA, calling for better

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articulation between methodology and linguistics courses in MA TESOL programs, and directly guiding candidates to apply their KAL to their ESL teaching.

3 Planning for language in content-based language instruction

Orientations toward planning for content and language instruction in a CBI classroom stem from diverse beliefs about form- versus meaning-focused instruction. For instance, teachers' comfort level and belief system regarding grammar instruction will bear upon their pedagogical choices and decisions to develop their own grammar foci for lessons, from 'abdicat[ing] grammar responsibilities to textbooks or materials supplied by others' (Andrews, 2001, p. 85), to having more form-based instructional approaches. These in turn relate to contextual influences, such as whether the instruction takes place in a single period of language instruction or as part of a total immersion program, and if the target language is learned in a highly scripted grammatical model or a content-based environment. Teachers who are focused on the content learning of students may neglect to attend to language form, letting errors go uncorrected and placing their emphasis on meaning rather than accuracy, while teachers who are oriented towards teaching language may provide much in the way of feedback to error and attention to language instruction, but struggle to situate this language in rich content learning (Cammarata, 2009; Fisher & Frey, 2010; Long, 2007). As Lyster (2007) sums up, 'content-based instruction that only alludes to language incidentally falls short of full-fledged integration, and decontextualized grammar instruction, by definition, precludes integration' (p. 26).

Lyster (2007), drawing primarily on research from Canadian French immersion classrooms, points out the need for a 'counterbalanced' approach. When teachers do not directly attend to language, language learners may not receive sufficient and extended language development necessary to achieve high levels of proficiency. He cautions that:

> much incidental attention to language is too brief and likely too perfunctory to convey sufficient information about certain grammatical subsystems and this, in those cases, can be considered neither systematic nor apt to make the most of content-based instruction as a means for teaching language. (p. 27)

Teachers' orientation towards form-focused instruction can be observed at the pre-active (lesson planning) stage, as Andrews (2001) states:

> TLA affects the teacher's ability to identify the key features of the grammar area for learning and to make them salient within the prepared input. It also affects the teacher's ability to specify the most appropriate learning objectives, and to select materials and tasks which are most likely to serve those objectives, ensuring that they are appropriate in terms of the learner's age, previous learning and present stage of interlingual development, and that they serve the desired learning outcomes. (p. 41)

4 Challenges for teachers setting language and content objectives in CBI

The ability to set clear student learning goals is widely appreciated as an important mental process for teachers to clarify their instructional plans (Marzano, Pickering & Pollock, 2001; Reeves, 2011). Underlying this is the belief that by formulating explicit objectives

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for learning, teacher candidates are more likely to develop instructional strategies and practices that will lead their students to what they need them to learn. In addition, since public schools regularly mandate that teachers set out their learning objective(s) for their students in each class, the focus on teacher candidates' abilities to do this for themselves fits squarely in line with what they will be expected to do in their future teaching. While the initial setting of objectives does not guarantee that teachers can apply in their instruction what they state are their intentions, nevertheless, objective setting is the starting point of this process. In CBI, setting both clear content and language objectives during the planning phase can be seen as one place where teachers must apply their explicit (declarative) TLA skills, and may be a means to ensure each receives focused attention during instruction (Hudson, Miller, & Butler, 2006; Short, Echevarria, & Richards-Tutor, 2011). The value of developing objectives is also supported by the findings of Echevarria, Short, and Powers (2006), who found that analysis of the language demands of the task, together with stated objectives for written and verbal language production, led to higher levels of achievement for ELLs. The nature of these language objectives has been the focus of investigation in a small number of studies, including Bigelow (2010), Bigelow and Ranney (2005), and Fisher and Frey (2010).

Fisher and Frey (2010) studied practicing mainstream teachers who had received their certification in California, where all teachers complete state licensing requirements for teaching ELLs through coursework related to English language development and ESL methodologies. Hundreds of language objective statements drawn from more than 300 K–12 participants were organized into three categories: those that focused on vocabulary, language structure, and language function. Teachers who instructed single-subject areas were the most likely to focus on discipline-specific vocabulary as their language objective (60%), and this was the most prevalent language focus (50% overall) in their data, which they found 'both understandable and problematic' (p. 330). About 30% were language objectives focusing on structure, such as using past tense verbs, of which about 15% involved the use of sentence frames. The final category of objectives focused on language functions (20%), e.g. justification, persuasion, and description. They also found that many of the language objectives submitted for their study were actually activities rather than learning purpose statements, suggesting that teachers may be unsure how their activities lead to specific language learning outcomes.

Bigelow (2010) and Bigelow and Ranney (2005) have enriched our understanding of the challenges for ESL teachers when confronted with the need to incorporate both content and language teaching goals within the already difficult job of lesson design. In her 2010 study, Bigelow investigated K–12 ESL teachers' choices and reflections on those choices as they planned language objectives within their content-based lesson plans, across a variety of practicum teaching settings. She found that in their planning, teachers mainly targeted vocabulary and grammar (mostly past tense verbs) objectives, and in teaching the skills, speaking and writing appeared more frequently than listening and reading. Elementary teachers also had many more form-focused language objectives than did secondary teachers.

In their 2005 study, Bigelow and Ranney found that regardless of where the teacher candidates began their planning (from the content or from the language goals), in the end the overall range of linguistic forms addressed across teachers' lesson plans was quite

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limited, with most attending to verb forms. While at first, as expected, teacher candidates focused on general timing and pacing issues in their planning, as they progressed through the practicum and gained more experience, they began to notice that the balance between content and language teaching objectives constituted a crucial struggle for them. In particular, developing language demands from authentic content material and teaching language objectives other than vocabulary were cited as difficulties. This was true especially when the texts, materials, or cooperating teacher did not make the language teaching point explicit, or when they felt they did not have sufficient metalinguistic knowledge or knowledge of students' language learning needs to identify language goals.

Figuring out how to focus on form, while breaking away from the strictly deductive, grammatical approach through which they had been taught a foreign language was another struggle for teachers, yet they were better able to contextualize a concrete language objective in content than they were able to do the reverse. 'In other words, beginning with content brought into play the classic struggle of CBI: The content often eclipses the language objectives of the lesson' (Bigelow & Ranney, 2005, p. 195).

III Method

Building upon the findings from these three most relevant studies, we employed a qualitative analysis of content and language objectives designed by TESOL teacher candidates, which were then reviewed through descriptive statistics. At our institution of higher education, candidates from five sections (n = 75) of the Student Teaching/Practicum course taken in Spring 2011 were invited to share four lesson plans submitted as part of routine classroom assignments. Each plan was actually utilized in the teacher candidate's classroom, and in all cases candidates were observed teaching from these plans. These plans were examined qualitatively by the authors to draft and then to validate a typology of issues in language objective writing. Results were analysed to determine salient patterns of clear and problematic objectives among different subject areas, grade levels, and language foci.

1 Participants

Participants for the study were students enrolled in the spring 2011 TESOL supervised teaching course in our Master's program leading to PreK–12 New York state teaching certification in TESOL, located in New York City. Students consisted of 'student teachers' (pre-service) and 'practicum teachers' (in-service). All candidates in these courses were invited to participate; none were excluded based on performance in coursework or for any other reasons. Student teachers were teaching for a 15-week semester in an elementary or secondary classroom under the guidance of a cooperating teacher. Practicum teachers were full-time teachers working under alternative licensing internship agreements. Approximately 65% of participants were practicum and 35% were student teachers.

This participant pool was used as we were best able to access the lesson plans and subsequent teaching observations from teacher candidates in our own institution. They also shared important characteristics. Practicum and student teachers take all of their

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classes together and were mixed together in each section of practicum. They also are equally represented in prior years of teaching experience: about one to three years. Because of alternative licensing routes to certification in our state, many true novices are employed while still studying and become 'in-service' teachers, while many student teachers have previously taught overseas or in private language schools, but are labeled 'pre-service'. We also have student teachers who have previous teaching certification in other subject areas, but in our state permanent teaching certification requires a masters degree, so they too are enrolled in our program. Finally, teacher candidates in the program are predominantly female (about 90%), ethnically diverse, most in their 20s or early 30s, and hold undergraduate degrees in various disciplines, but primarily in the humanities and social sciences. For these reasons, we included both student and practicum teachers in their final semester of supervised teaching as participants, but would point out that in this sample, the majority of the lesson plans were submitted from in-service teachers (65%). One limitation to the sample used is that we were not able to track the submitted lesson plans back to the individual; thus, we were not able to run inferential statistics linking their background variables to their plans. We wanted to assure participants anonymity as we are faculty in their TESOL program and as part of the Institutional Review Board process, protection and avoidance of coercion was paramount.

2 Materials

Materials consisted of teacher candidates' lesson plans from the programs' practicum courses. Lesson plans from 43 candidates were used for the final analyses, with 15 other candidates' plans utilized in the exploratory phases of the project. The lesson plans followed a standard template required by all instructors in the program. This template had been developed with input from faculty within TESOL and across content disciplines in the college as an instructional support, to be used through out the MA TESOL program from initial lesson plan exercises to subsequent use in their practicum teaching. In length and complexity this lesson plan template was conceived of as a guide to develop candidates' pedagogical reasoning while in their MA course.

We used 15 plans initially in an exploratory phase of the study (Baecher & Farnsworth, 2011). For the validation of and final adjustments to the taxonomy, a total of 107 plans were analysed from 43 teacher candidates, of which 69 plans were for elementary and 38 for secondary grades. In terms of content area, 73 plans, or 68% of the total, were for English Language Arts (ELA) content, nine were for mathematics, seven for science, and 18 for social studies. Table 1 presents the number of plans by subject area and grade level.

3 Procedures

Data analysis was conducted in two phases. First, in an exploratory phase, 15 lesson plans were examined and discussed in order to arrive at a taxonomy of issues and descriptors of clear lesson objectives. Second, in a validation phase, 107 plans were examined using the taxonomy to further test and refine the instrument and to analyse the frequency of various problematic issues in the plans.

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Table 1. Number of plans by grade and subject area.

Subject	Number of elementary lesson plans	Percentage total	Number of secondary lesson plans	Percentage total	Combined totals	Combined percentages
ELA	43	40	30	28	73	68
Mathematics	8	8	1	1	9	9
Science	7	7	0	0	7	7
Social Studies	11	10	7	6	18	16
Totals	69	65	38	35	107	100

The first 15 plans were examined by all three authors in multiple meetings. The purpose of these meetings was, first, to agree upon what constituted clear and high quality content and language objectives. We agreed on definitions and chose model plans that exemplified our definitions. These definitions, along with example objectives, constituted our typology. Our definition of clear content area objectives was as follows:

1. The objective specifies the content area (e.g. science, social studies, ELA, math) understanding that students will achieve by the end of the lesson.
2. Lesson plans with focused content objectives have the following characteristics: (a) one or more specific conceptual, cognitive, or topical understandings that learners could arrive at in a period of instruction; (b) specific activities to either expose, introduce, and/or provide guided or freer practice to develop this understanding.

Next we examined plans we determined to be problematic, in that they did not meet the definition above in some noteworthy respect. This was a qualitative judgment, driven by the three authors' experiences and multiple consensus-building meetings held over the course of 8 months. We further examined the problematic plans and constructed a typology of common problem areas we saw within these plans. We created descriptors for each type of problem, with some problems common to both content and language objectives and some unique to language objectives. The descriptors included examples drawn from the lesson plans. The following categories were established to describe problematic content area objectives:

- Vaguely/broadly worded: These objectives were unclear due to vagueness or overly broad descriptions of the understandings they described. For example, one content objective stated: 'Students will use the scientific method.'
- Undigested standard: Some objectives merely quoted verbatim a portion of the state content area learning standards, which while useful do not in general describe understandings at the lesson level of achievement. For example, one ELA content objective stated, 'Students will be able to read, write, listen, and speak for information and understanding, literary response and expression, critical analysis and evaluation, and social interaction.'

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- Not feasible in a single period of instruction: Some objectives were clearly not feasible within a single period of instruction, which was at the core of our definition. For example, one objective stated, 'Students will understand the themes present in the novel *The Lovely Bones*.'
- Describes an activity rather than a learning goal: The objectives described what students would do during the lesson instead of what they would learn. For example, one objective stated, 'Students will create their own graphic organizer, after reading a short biography, about the early years of their inventors' lives.'

The lesson plans were then examined to determine a definition for clear language objectives within the content lesson, namely:

1. The objective specifies the language knowledge, specific to the L2 learner, and the ability to use it that students will achieve by the end of the lesson.
2. Lesson plans with focused language objectives have the following characteristics: (a) one or more specific language functions, grammatical structures, microskills, learning strategies, or vocabulary that learners will learn about and be able to use in a period of instruction; (b) specific activities to either expose, introduce, and/or provide guided or freer practice to develop this understanding or skill.

The following categories were established for problematic language objectives. The first four categories mirrored those for the content objectives:

- Vaguely/broadly worded: These objectives were unclear due to vagueness or overly broad descriptions of the understandings they described. For example, one language objective stated: 'Students will develop listening skills.'
- Undigested standard: Some objectives quoted verbatim a portion of the state ESL standards, which do not in general describe understandings at the lesson level of achievement, nor does this approach demonstrate pedagogical reasoning. For example, 'Students will read, write, listen and speak for communication.' This sentence is a descriptor from the New York State ESL standards.
- Not feasible in a single period of instruction: Some objectives were clearly not feasible within a single unit of instruction. For example, one lesson had as its language objective, 'Students will make and support inferences about information and ideas with reference to features in oral and written text. Such features include vocabulary, format, facts, sequence, register, and relevance of details.'
- Describes an activity rather than a learning goal: The objectives described what students would do during the lesson instead of what they would learn. For example, one lesson stated: 'Students will describe pictures and situations using the present progressive tense.'
- Describes an ELA rather than an ESL goal, where it should be clearly distinct: The objectives did not include any understandings which would be unique to or particularly appropriate for ESL learners as opposed to general education students. For example, one lesson objective stated: 'Students will understand that authors write with different purposes.'

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- Clear mismatch between content objectives and language objectives: The language objectives bore no logical or functional relationship to the content objectives. For example, one language objective stated: 'Students will identify parts of speech', whereas the content objective stated: 'Students will understand why Beowolf is characterized as an epic hero.'

In addition to analysing the objectives, we categorized the specific language focus of the plans by establishing definitions for the following foci: (1) vocabulary, (2) grammar (and the specific grammar focus), (3) language function, (4) language strategy use, (5) subskill of reading, writing, listening or speaking (and what subskill), and (6) sentence frame or starter. Sentence starters are beginnings of statements to scaffold speaking or writing activities, such as 'I agree with _____ because …', while a sentence frame is a larger text with words omitted, both provided by the teacher to scaffold the language activity. We viewed these as an indication that the teacher was considering language development, even if the objective was not specific to a language form. We categorized the plans along these lines regardless of whether the language and/or content objectives had been rated as clear.

Procedures and guidelines for rating were discussed and agreed upon. To some extent this was an iterative process, reflecting the exploratory nature of the research. We decided that even if the lesson activities reflected feasible objectives for content or language, it was the formal statement of objectives we would evaluate primarily. Additionally, if the objective implied that the target language objective was being developed over a period of time, as in 'Students will be able to begin to notice that word final *-s* has a variety of meanings,' that counted as a valid language objective, recognizing that language skills are not developed in a single period, but over time. Also, we decided to use caution in judging whether an objective was trivial or inappropriate for a given group of learners, because in most cases we did not know the particulars of the students or learning environment. This aspect of judging the quality of objectives – the degree of fit between the objectives and the particular learners' needs – was particularly difficult.

At least two of the authors examined and categorized each plan independently using the typology described above and, in cases of disagreement, a third author examined it as well and, through process of discussion, we arrived at a consensus categorization for each plan. This consensus model precluded an analysis of inter-rater reliability, but given the nature of the research and the variety of lesson plans encountered, it was deemed appropriate. We looked for salient patterns of both clear and problematic objectives among the content and language objectives.

IV Findings

Analysis of the 107 lesson plans revealed a wide variety of challenges for teacher candidates. First we describe overall trends among the clear content and language objectives, and then we discuss tendencies among the problematic objectives.

The majority of the plans analysed had an ELA content area focus. Within content objectives, ELA objectives were clear less often than those of other subjects. We found that 47% of the ELA objectives were clear versus 74% for the other objectives combined.

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Table 2. Clear content objectives by grade and subject area (data in parentheses is the *n* of the population in the given dataset).

Subject	Number of elementary lesson plans	Percentage total	Number of secondary lesson plans	Percentage total	Combined totals	Combined percentages
ELA	22 (43)	51	12 (30)	40	34 (73)	47
Mathematics	8 (8)	100	1 (1)	100	9 (9)	100
Science	5 (7)	71	0 (0)	n/a	5 (7)	71
Social Studies	9 (11)	82	2 (7)	29	11 (18)	61
Totals	44 (69)	64	15 (38)	39	59 (107)	55

Table 3. Clear language objectives by grade and subject area (data in parentheses is the *n* of the population in the given dataset).

Subject	Number of elementary lesson plans	Percentage clear elementary	Number of secondary lesson plans	Percentage clear secondary	Totals	Percentage total
ELA	21 (43)	49	8 (30)	27	29 (73)	40
Mathematics	3 (8)	38	0 (1)	0	3 (9)	33
Science	4 (7)	57	0 (0)	0	4 (7)	57
Social Studies	4 (11)	36	1 (7)	17	5 (18)	28
Totals	32 (69)	46	9 (38)	24	41 (107)	38

This may indicate that ELA content-based lesson plans pose unique challenges for TESOL teacher candidates. Content objectives were judged as clear more often than language objectives were. As to the content objectives, 55% of them were rated as clear, as opposed to only 38% of the language objectives. Interpreting this result is difficult; it may indicate that our students indeed struggle more with language objectives than with content objectives. However we, as language specialists, may also have been more severe in our ratings of language objectives. Table 2 provides a summary of the content area objectives rated as acceptable, by subject area and grade level.

Objectives for elementary grades were rated satisfactory more often than those for secondary grades, for both language and content. Overall, the number of satisfactory language objectives was low, raising questions about our teaching of this important skill. The number of satisfactory language objectives in mathematics and in social studies was only about 30% combined, which may indicate that crafting language objectives in these content domains is challenging for our students. Table 3 provides a summary of the language objectives rated as acceptable, by subject area and grade level.

Next, we examined the types of problems encountered among the unsatisfactory content area objectives. In the elementary grades, 25 lesson plans were classified as problematic, while in the secondary grade plans, 23 were problematic. Some plans were rated problematic in more than one area; therefore, more problems were identified than there were plans. Within the elementary context, about half of the content objectives were

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Table 4. Problematic language objectives by grade (percentages in parentheses).

Problem	Elementary lesson plans	Secondary lesson plans	Totals
Too Vague	20 (33)	11 (21)	31 (27)
Undigested Standard	12 (19)	5 (9)	17 (15)
Not Feasible	6 (10)	6 (12)	12 (11)
Activity, not Goal	12 (20)	14 (27)	26 (23)
ELA, not ESL	7 (11)	14 (27)	21 (19)
Mismatch between Objective and Instruction	4 (7)	2 (4)	6 (5)
Totals	61 (100)	52 (100)	113 (100)

Table 5. Foci of language objectives by grade.

Language focus	Elementary lesson plans	Secondary lesson plans	Totals
Vocabulary	19 (25)	7 (23)	26 (25)
Structure	12 (16)	6 (20)	18 (17)
Function	4 (5)	3 (10)	7 (6)
Strategy	0 (0)	2 (7)	2 (2)
Subskill	30 (39)	10 (33)	40 (38)
Sentence Frame	11 (14)	2 (7)	13 (12)
Totals	76 (100)	30 (100)	106 (100)

evaluated as problematic from either being too vague (10) or being constructed as activities rather than learning goals (13). In contrast, among the secondary lesson plans, very few content objectives were written as activities (4), but about a third were written as undigested standards (9) or appeared not to be feasible (9) in our sample.

We then examined the types of problems encountered in the language objectives rated unsatisfactory. In the elementary grades, 37 language objectives were problematic, while among the secondary plans, 29 language objectives were problematic. Some plans were problematic in more than one area. Table 4 below provides the values for the various problems identified, by grade and type of problem.

Again, there were some noteworthy patterns. In secondary grades, ELA not ESL (the objective had no connection to ESL instruction specifically) was a bigger problem than it was in elementary grades. We noted that the 'activity, not goal' problem – where language objectives describe classroom activities to be completed rather than student competencies or understandings to be addressed – was persistent across grades and subject areas. So, too, was the problem of 'too vague' objectives. Of the plans that were problematic in multiple areas (more common with language objectives), the most common problems were 'too vague' and 'ELA not ESL' in combination with other problems.

Finally, we analysed the language objectives further to determine if they focused on vocabulary, structure, function, strategy, subskill within the four skills, such as 'using context clues in reading', or sentence frames. Table 5 provides an overview of the

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language foci found in the plans by type of focus and grade level. Note that some lesson plans had no discernible focus, and some had more than one.

As Table 5 shows, lesson plans most commonly focused on one of the subskills of reading, writing, listening, or speaking than on any other area of language. The second most common focus was vocabulary for both elementary and secondary grades. 'Sentence frame' objectives were common in elementary grades but less so in secondary grades. Overall there was little focus on language function or strategy use as we defined them.

Finally, we further analysed each of the language categories. When grammar was a focus, it was primarily on verb tenses (simple present and past) or adjectives. The only other structures featured – several more than once – were transition words (either sequence words or comparison/contrast words), identifying parts of speech (noun, verb, adjective, adverb only), and sentence structure (involving peer correction of sentences in writing). When specific language skills/modes were focused on, reading and writing were most commonly addressed, with 14 plans focusing on some aspect of writing and 14 on reading comprehension. Of the remaining plans, six focused on speaking skills, and four on listening.

V Discussion

Supporting the findings of Bigelow (2010), Bigelow and Ranney (2005), and Fisher and Frey (2010) results of our investigation suggest that TESOL teacher candidates vary in their ability to draft clear content and language objectives, according to the criteria that we developed. Participants in our sample tended to write clear content objectives. However, they were less successful at developing language objectives. The language objectives they did develop focused most heavily on the four language skills, next most often on vocabulary, and considerably less often on grammatical structures, functions, or language learning strategies.

Our findings indicated that the teacher candidates in our data experienced several clear difficulties in how they tended to craft language objectives. The first was difficulty focusing down to the level of a single lesson for their objective, instead confusing language objectives with higher-level expectations of ESL students (undiluted standards statements), or with far too broad, undefined general notions (which we defined as vague, or unfeasible). This may have occurred due to a lack of understanding about what a language objective is, because they also confused it with an activity, or they struggled with identifying what sort of language focus might fit with particular content material. The tendency for some of the lesson plans to have multiple problems – generally, vague or unfeasible, plus another difficulty – suggests that many teacher candidates need additional work understanding what a language objective is and how specific it needs to be. This may also be a characteristic of applying TLA knowledge to lesson planning among novice teachers, as they lack the experience necessary to successfully visualize the appropriate scope of objectives, resulting in objectives that are written too broadly rather than narrowly.

A second general area of confusion that appeared equated language foci with what is often taught in general ELA instruction (as opposed to ESL instruction), a distinction that is, even for researchers, difficult to make. In our view of 'content', we defined ELA

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content as involving instruction in literacy skills (reading, writing, etc.), as well as literary analysis and appreciation of various genres of fiction and non-fiction. While ESL instruction also focuses on instruction in the various skills, it must be distinguished from ELA by the fact that it includes specific instruction on the language needed to develop the above literacy skills. For example, in both ELA and ESL, as part of teaching a child to read, a teacher might focus on prediction as a strategy to help develop better comprehension. While both ELA and ESL classes might address the idea that we use information we can glean from briefly perusing a book in order to predict what it might be about (thereby activating more of the students' world knowledge and facilitating their comprehension), what distinguishes an ELA objective from an ESL objective might be that an ELA class for non-ELLs would not have to include instruction on the specific language ESL students might need to know first in order to be able to express a prediction using such expressions as 'I think I will find …' or 'I predict that X will happen …'

Third, teacher candidates tended to focus on language sub-skills or vocabulary more often than structure, function, or other possible foci. The strong focus on subskills could also be explained as an area of overlap between ELA and ESL instruction as explained above, an issue that possibly confounded our findings as well. This overlap is not surprising, given that it is common for ESL instruction to focus on reading, writing, and other skill areas that also are commonly considered to be within the subject area of ELA literacy instruction, particularly at the elementary level. The tendency to focus on language sub-skills typical of ELA instruction may also be a function of limited time in a single lesson to address both content and language goals. The tendency to focus on vocabulary could be explained as one area of language that teachers in general, particularly those not trained to teach ELLs, would be most likely to focus on. Since a number of the participants in this study have previous certification and teaching experience in general education classes (particularly in elementary education), a focus on vocabulary would already be a familiar exercise to them. Similarly, vocabulary-based ESL objectives may also be the type they have most frequently witnessed in CBI classrooms. However, the lack of focus on other aspects of language (grammar, language functions, etc.) also suggests that teacher candidates may not be receiving adequate or appropriate focus on language analysis in their preparation.

Fourth, the narrow range of grammatical features addressed in their language objectives could be explained in several ways. First, as Bigelow & Ranney (2005) also found, it could indicate that they were not sufficiently exposed to ways of teaching CBI lessons with a focus on form (Doughty & Williams, 1998). Bigelow and Ranney (2005) suggest that teacher candidates may be unable to teach grammar in ways that do not match their own preconceived notions of how grammar should be taught, either with grammar not taught at all, or traditionally in decontextualized lessons focused solely on grammar structures. A second possible explanation for teacher candidates' limited focus on grammar may result from a lack of confidence in selecting such objectives for lessons for which they would subsequently be observed. Since teacher candidates will tend to avoid risk-taking in practicum observations (Richards, 1996), they may have chosen language objectives they felt more comfortable teaching. A third explanation is that their avoidance may be due to their perceived or genuinely inadequate preparation in this area, a finding suggested by others as well (e.g. Ediger & McCormack, 2007).

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Ediger and McCormack investigated a slightly different, but related aspect of teacher candidates' declarative TLA, identifying and designing instruction for remedying ESL student writing errors. Parallel to the current study, they found that teacher candidates had limited success in identifying the structures involved in their students' errors, but even less success designing appropriate instruction to remedy those errors. Finally, two other possible explanations for limited focus on grammar are that some of our teacher candidates may have been influenced by recent tendencies not to focus on explicit grammar instruction in some US public schools, a possible effect of recent trends in communicative or 'naturalistic' language teaching (Andrews, 2007), or because of a belief that children and other low-level learners are not ready for explicit grammar instruction. Graduate school programs may need to address more directly teacher candidates' internal conflicts between a school's required or prohibited practices regarding explicit grammar instruction and their graduate school course content that tells them that such focus on form within meaningful CBI contexts is the research-supported methodology they should follow.

VI Conclusions and recommendations for teacher educators

Like Mutton, Hagger and Burn (2011), we conclude from our investigation that 'the expectation that student teachers will arrive in school having been taught how to plan at the university level is clearly a fallacious one' (p. 412), and this is perhaps especially so when novice teachers are faced with the dual challenge of planning content and language goals. Our findings suggest that instruction on designing language objectives as one aspect of their preparation for teaching – whether in methods, pedagogical grammar, or practicum courses – cannot assume teacher candidates' automatic ability to convert TLA knowledge into the skill of designing such objectives for specific lessons, for specific groups of ELLs, within the new meaningful content they are simultaneously expected to teach. Implications for our MA TESOL program, which may be relevant to other TESOL preparation programs, include a need for:

1. greater mentoring of and focus on the application of TLA knowledge through collaborative lesson planning, especially during the practicum, at which time teacher candidates will be able to begin to draw upon knowledge of real pupils and be able to see how their cooperating teacher plans for instruction;
2. more explicit modeling of:
    a. how to create language objectives focusing on different areas of language (a wider array of grammar structures, functions, discourse, etc.) within meaningful contexts, along with increased focus on the building of teacher candidates' confidence in their ability to design instruction in these areas;
    b. how to identify and pull language foci out of particular content material;
    c. which language might specifically be required in order to understand, discuss, or write about specific content, perhaps with faculty conducting 'think-aloud' sessions to unpack the lesson planning process;

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3. a greater focus on developing teacher candidates' ability to design ELA content objectives in particular, with time spent discussing overlapping areas and differences between ELA and ESL objectives;
4. the need for change in our teacher candidates' notions of the usefulness and appropriateness of explicit grammar/language instruction; and
5. Addressing the design of learning objectives and other applications of TLA in multiple places in the TESOL curriculum, recycling and spiraling this knowledge and skill across the program.

Future research could investigate the implementation of such approaches and their effect on the quality of teachers' language objectives; collecting observational data from actual lessons taught in PreK–12 classrooms and their relationship to the language objectives set in the plans for those lessons; and tracking teachers over time to understand more about how they operationalize content and language planning as they move from abstract understandings of language to applying it to the design and execution of instruction. Concerns about the integration of content and language planning will continue to be of great relevance to all teachers of ELLs in content-based classrooms, and those involved in teacher preparation will continue to benefit from a greater understanding of teacher candidates' struggles in providing targeted English language development in CBI.

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References

Andrews, S. (2001). The language awareness of the L2 teacher: Its impact upon pedagogical practice. *Language Awareness*, *10*, 75–90.

Andrews, S. (2003). Teacher language awareness and the professional knowledge base of the L2 teacher. *Language Awareness*, *12*, 81–95.

Andrews, S. (2007). *Teacher language awareness*. Cambridge: Cambridge University Press.

Arkoudis, S. (2006). Negotiating the rough ground between ESL and mainstream teachers. *International Journal of Bilingual Education & Bilingualism*, *9*, 415–433.

Baecher, L., & Farnsworth, T. (2011). Understanding teachers' challenges in writing language objectives in content-based instruction. Paper presented at the Language Teacher Education Conference, Minneapolis, MN, USA, May, 2011.

Bailey, K. (1996). The best laid plans: Teachers' in-class decisions to depart from their lesson plans. In K. Bailey & D. Nunan (Eds.), *Voices from the language classroom: Qualitative research in second language education* (pp. 15–40). Cambridge: Cambridge University Press.

Bigelow, M. (2010). Learning to plan for a focus on form in CBI: The role of teacher knowledge and teaching context. In J. Davis (Ed.), *World language teacher education: Transitions and challenges in the twenty-first century* (pp. 35–56). Charlotte, NC: Information Age Publishing.

Bigelow, M., & Ranney, S. (2005). Pre-service ESL teachers' knowledge about language and its transfer to lesson planning. In N. Bartels (Ed.), *Applied linguistics and language teacher education: Volume 4* (pp. 179–200). New York: Springer Science/ Business Media.

Bullough, R.V. (1992). Beginning teacher curriculum decision making, personal teaching metaphors, and teacher education. *Teaching and Teacher Education*, *8*, 239–252.

Downloaded from ltr.sagepub.com by guest on January 7, 2016

Cammarata, L. (2009). Negotiating curricular transitions: Foreign language teachers' learning experience with content-based instruction. *Canadian Modern Language Review/ La Revue Canadienne Des Langues Vivantes*, *65*, 559–585.

Creese, A. (2006). Supporting talk? Partnership teachers in classroom interaction. *International Journal of Bilingual Education & Bilingualism*, *9*, 434–453.

Dalton-Puffer, C. (2007). *Discourse in content and language integrated learning (CLIL) classrooms*. Amsterdam: John Benjamins.

Doughty, C., & Williams, J. (1998). *Focus on form in classroom SLA*. Cambridge: Cambridge University Press.

Dunn, T.G., & Shriner, C. (1999). Deliberate practice in teaching: What teachers do for self-improvement. *Teaching and Teacher Education*, *15*, 631–651.

Echevarria, J., Short, D., & Powers, K. (2008). Making content comprehensible for non-native speakers of English: The SIOP Model. *The International Journal of Learning*, *14*, 41–50.

Ediger, A., & McCormack, B. (2007). Grammar knowledge requirements in ESL teacher preparation as evidenced through novice teachers' written feedback on lexical and grammatical errors. Paper presented at the American Association of Applied Linguistics (AAAL) Conference, Costa Mesa, CA, April 2007.

Fisher, D., & Frey, N. (2010). Unpacking the language purpose: Vocabulary, structure, and function. *TESOL Journal*, *1*, 315–337.

Gándara, P., Maxwell-Jolly, J., & Driscoll, A. (2005). *Listening to teachers of English language learners: A survey of California teachers' challenges, experiences, and professional development needs*. Berkeley, CA: Policy Analysis for California Education.

Ho, B. (1995). Using lesson plans as a means of reflection. *ELT Journal*, *49*, 66–71.

Hudson, P., Miller, S.P., & Butler, F. (2006). Adapting and merging explicit instruction within reform based mathematics classrooms. *American Secondary Education*, *35*, 19–32.

John, P.D. (1991). Course, curricular and classroom influences on the development of student teachers' lesson planning perspectives. *Teaching and Teacher Education*, *7*, 359–373.

John, P.D. (1994). The integration of research validated knowledge with practice: Lesson planning and the student history teacher. *Cambridge Journal of Education*, *24*, 33–49.

John, P.D. (2006). Lesson planning and the student teacher: Re-thinking the dominant model. *Journal of Curriculum Studies*, *38*, 483–498.

Kagan, D., & Tippins, D.J. (1992). The evolution of functional lesson plans among twelve elementary and secondary student teachers. *The Elementary School Journal*, *92*, 477–491.

Liyanage, I., & Bartlett, B.J. (2010). From autopsy to biopsy: A metacognitive view of lesson planning and teacher trainees in ELT. *Teaching and Teacher Education*, *26*, 1362–1371.

Long, M. (2007). *Problems in Second Language Acquisition*. Mahwah, NJ: Lawrence Erlbaum.

Lyster, R. (2007). *Learning and teaching languages through content: A counterbalanced approach*. Amsterdam: John Benjamins.

Mackey, A., Polio, C., & McDonough, K. (2004). The relationship between experience, education, and teachers' use of incidental focus on form techniques. *Language Teaching Research*, *8*, 301–327.

Marzano, R.J., Pickering, D.J., & Pollock, J.E. (2001). *Classroom instruction that works: Research-based strategies for increasing student achievement*. Alexandria, VA: Association for Supervision and Curriculum Development.

McCutcheon, G., & Milner, H. R. (2002). A contemporary study of teacher planning in a high school English class. *Teachers and Teaching: Theory and Practice*, *8*, 81–94.

Menken, K., & Antunez, B. (2001). *An overview of the preparation and certification of teachers working with limited English proficient (LEP) students*. Washington, DC: National Clearinghouse for Bilingual Education.

Downloaded from ltr.sagepub.com by guest on January 7, 2016

Met, M. (1991). Learning language through content: Learning content through language. *Foreign Language Annals*, *24*, 281–295.

Met, M. (1999). *Content-based instruction: Defining terms, making decisions*. Washington, DC: The National Foreign Language Center. Retrieved from: http://www.carla.umn.edu/cobaltt/modules/principles/decisions.html (September 2013).

Mutton, T., Hagger, H., & Burn, K. (2011). Learning to plan, planning to learn: The developing expertise of beginning teachers. *Teachers and Teaching*, *17*, 399–416.

Nunan, D. (1992). *Research methods in language learning*. Cambridge: Cambridge University Press.

Pica, T. (2008). Task-based instruction. In N. Van Deusen-Scholl, & N.H. Hornberger (Eds.), *Encyclopedia of language and education: Volume 4: Second and Foreign Language Education*. 2nd ed. (pp. 71–82). New York: Springer Science/ Business Media.

Popko, J. (2005). How MA-TESOL students use knowledge about language in teaching ESL classes. In N. Bartels (Ed.), *Applied linguistics and language teacher education: Volume 4* (pp. 387–403). New York: Springer Science/ Business Media.

Reeves, A.R. (2011). *Where great teaching begins: Planning for student thinking and learning*. Alexandria, VA: Association for Supervision and Curriculum Development.

Richards, J.C. (1996). Teachers' maxims in language teaching. *TESOL Quarterly*, *30*, 281–296.

Richards, J.C., & Pennington, M.C. (1998). The first year of teaching. In J.C. Richards (Ed.), *Beyond training* (pp. 173–190). Cambridge: Cambridge University Press.

Sardo-Brown, D. (1996). A longitudinal study of novice secondary teachers' planning: Year two. *Teaching and Teacher Education*, *12*, 519–530.

Short, D. (2002). Language learning in sheltered social studies classes. *TESOL Journal*, *11*, 18–24.

Short, D., Echevarria, J., & Richards-Tutor, C. (2011). Research on academic literacy development in sheltered instruction classrooms. *Language Teaching Research*, *15*, 363–380.

Shulman, L. (1986). Those who understand: Knowledge growth in teaching. *Educational Researcher*, *15*, 4–14.

Stoller, F.L. (2004). Content-based instruction: Perspectives on curriculum planning. *Annual Review of Applied Linguistics*, *24*, 261–283.

Stoller, F.L. (2008). Content-based instruction. In N. Van Deusen-Scholl & N.H. Hornberger (Eds.), *Encyclopedia of language and education: Volume 4: Second and Foreign Language Education*. 2nd ed. (pp. 59–70). New York: Springer Science/ Business Media.

Stoller, F.L., & Grabe, W. (1997). A six-T's approach to content-based instruction. In M.A. Snow & D. Brinton (Eds.), *The content-based classroom* (pp. 78–94). New York: Longman.

Warren, L.L. (2000). Teacher planning: A literature review. *Educational Research Quarterly*, *24*, 37–42.

Westerman, D.A. (1991). Expert and novice teacher decision making. *Journal of Teacher Education*, *42*, 292–305.

Wiggins, G., & McTighe, J. (2005). *Understanding by design*. 2nd ed. Alexandria, VA: Association for Supervision and Curriculum Development.

Williamson, J., & Hardman, F. (1995). Time for refilling the bath?: A study of primary student-teachers' grammatical knowledge. *Language and Education*, *9*, 117–134.

Woods, D. (1996). *Teacher cognition in language teaching: Beliefs, decision-making and classroom practice*. Cambridge: Cambridge University Press.

Yildirim, A. (2003). Instructional planning in a centralized school system: Lessons of a study among primary school teachers in Turkey. *International Review of Education*, *49*, 525–543.

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